

Rotation Formulas:

$$1) \text{ Find } \theta \text{ where } \cot 2\theta = \frac{a-c}{b} = \frac{0-0}{1} = 0$$

$$2\theta = \frac{\pi}{2} = 0$$

$$\theta = \frac{\pi}{4}$$

$$\cos\left(\frac{\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$$

Substitute into xy : $x = x_2 \cos \theta - y_2 \sin \theta$, $y = x_2 \sin \theta + y_2 \cos \theta$

$$= \frac{x_2 - y_2}{\sqrt{2}}$$

$$= \frac{x_2 + y_2}{\sqrt{2}}$$

$$xy = \frac{(x_2 - y_2)(x_2 + y_2)}{2} = \frac{x_2^2 - y_2^2}{2}$$

Substitute into original equation:

$$xy + 7 = 0 \rightarrow \frac{x_2^2 - y_2^2}{2} + 7 = 0$$

$$x_2^2 - y_2^2 + 14 = 0$$

$$x_2^2 - y_2^2 = -14$$

$$y_2^2 - x_2^2 = 14$$